

PES University — Dept. of CSE

Lab 1: Requirements Table — Community Solar Credit Allocation Manager (Problem Statement #63)

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall record solar generation units (kWh) from smart meter feeds and allow prosumers to transfer surplus credits to neighbor accounts.	High	Pass: Transferred credits deduct from prosumer balance and apply to neighbor billing statement. Fail: Transfer exceeding available balance is rejected.	Core mechanism enabling peer-to-peer solar credit sharing within the microgrid.
FR-002	Functional	The system shall display each prosumer's real-time solar generation total, current credit balance, and monthly billing offset on a personal dashboard.	High	Pass: Dashboard values match the backend ledger within 5 minutes of the last smart-meter update. Fail: Dashboard shows stale or mismatched values.	Gives residents the visibility needed to decide when and how much surplus credit to share.
FR-003	Functional	The system shall allow a prosumer to submit a request to transfer a specified number of surplus credits to a chosen neighbor's account, validating that the requested amount does not exceed the prosumer's available balance before processing.	High	Pass: Requests within the available balance are processed and confirmed to both parties. Fail: Requests exceeding balance are rejected with an on-screen error and no ledger change occurs.	Prevents overdrawn transfers; realizes the «include» balance-validation step (UC-02 includes UC-03).
FR-004	Functional	The system shall automatically flag any single credit transfer exceeding a configurable threshold (e.g., 500 kWh) for review and approval by the Co-op Manager before the credits are applied to the recipient's account.	Medium	Pass: Transfers above the threshold are held in a 'Pending Approval' state and applied only after manager approval. Fail: A transfer above the threshold is applied without manager sign-off.	Mitigates fraud/error risk and supports fair allocation of shared microgrid capacity; realizes the «extend» relationship (UC-06 extends UC-02).
FR-005	Functional	The system shall generate a monthly billing offset report, per household, summarizing kWh generated, credits transferred, credits received, and net utility bill offset, exportable for Co-op	Medium	Pass: Report totals reconcile exactly with the sum of individual household ledger entries for the month.	Required for accurate utility billing reconciliation and co-op transparency.

		Manager review and utility submission.		Fail: Report omits a household or totals do not reconcile.	
NF R-001	Nonfunctional Performance & Security	The smart meter telemetry processing pipeline must validate and store hourly generation data for 500 households with 99.9% uptime.	High	Pass: Benchmarking tests confirm target latency and security standards are met under simulated peak load.	Ensures the data-ingestion layer scales reliably to the full neighborhood microgrid.
NF R-002	Nonfunctional Security & Privacy	The system shall encrypt all personally identifiable and billing data at rest (AES-256) and in transit (TLS 1.2+), and restrict access to household-level generation and credit data to the account owner and authorized Co-op Manager roles only.	High	Pass: A security audit/penetration test confirms no PII is stored or transmitted unencrypted, and role-based access controls block cross-household data access. Fail: Any PII is found unencrypted, or a non-owner/non-manager role can view another household's data.	Solar generation and billing data are sensitive household financial data; protecting them is essential for resident trust and regulatory compliance.